

From Time Dilation to Mass Variation:

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Time field connection $\Gamma_\mu^{(T)} = -\partial_\mu m/m^2 \rightarrow -\partial_\mu m/m$; note on the dimensional check of $\mathcal{L}_{\text{time}}$ ($[E^{-2}]$ and $[E^0]$ differ, check mark not justified).
- (1 Oct. 2026) Higgs check relation: "uniquely determined", $\xi \approx 1.33 \times 10^{-4}$ with $\lambda_h \approx 0.13 \rightarrow$ consistency check, $\approx 1.30 \times 10^{-4}$ with $\lambda_h = m_h^2/(2v^2) \approx 0.129$ (about 2.4 % below 4/30000); $\beta_T = 1$ "uniquely determines ξ " $\rightarrow$ a setting, with SM values $\beta_T \approx 0.976$ (Doc. 385).

Abstract

This updated work presents the essential mathematical formulations of time-mass duality theory, building upon the comprehensive geometric foundations established in the field-theoretic derivation of the β parameter. The theory establishes a duality between two complementary descriptions of reality: the standard view with time dilation and constant rest mass, and the T0 model with absolute time and variable mass. Central to this framework is the intrinsic time field $T(x, t) = \frac{1}{\max(m, \omega)}$ (in natural units where $\hbar = c = \alpha_{\text{EM}} = \beta_T = 1$), which enables a unified treatment of massive particles and photons through the three fundamental field geometries: localized spherical, localized non-spherical, and infinite homogeneous. The mathematical formulations include complete Lagrangian densities with strict dimensional consistency, incorporating the derived parameters $\beta = 2Gm/r$, $\xi = 2\sqrt{G} \cdot m$, and the cosmic screening factor $\xi_{\text{eff}} = \xi/2$ for infinite fields. All equations maintain perfect dimensional consistency and contain no adjustable parameters.

.1 Introduction: Updated T0 Model Foundations

This updated mathematical formulation builds upon the comprehensive field-theoretic foundation established in the T0 model reference framework. The time-mass duality theory now incorporates the complete geometric derivations and natural units system that demonstrate the fundamental unity of quantum and gravitational phenomena.

Fundamental Postulate: Intrinsic Time Field

The T0 model is based on the fundamental relationship between time and mass expressed through the intrinsic time field:

$$T(x, t) = \frac{1}{\max(m(x, t), \omega)} \quad (1)$$

Dimensional verification: $[T(x, t)] = [1/E] = [E^{-1}]$ in natural units ✓

This field satisfies the fundamental field equation derived from geometric principles:

$$\nabla^2 m(x, t) = 4\pi G \rho(x, t) \cdot m(x, t) \quad (2)$$

Dimensional verification: $[\nabla^2 m] = [E^2][E] = [E^3]$ and $[4\pi G \rho m] = [1][E^{-2}][E^4][E] = [E^3]$ ✓

Three Fundamental Field Geometries

The complete T0 framework recognizes three distinct field geometries with specific parameter modifications:

T0 Model Parameter Framework

Localized Spherical Fields:

$$\beta = \frac{2Gm}{r} \quad [1] \quad (3)$$

$$\xi = 2\sqrt{G} \cdot m \quad [1] \quad (4)$$

$$T(r) = \frac{1}{m_0}(1 - \beta) \quad (5)$$

Localized Non-spherical Fields:

$$\beta_{ij} = \frac{r_{0ij}}{r} \quad (\text{tensor}) \quad (6)$$

$$\xi_{ij} = 2\sqrt{G} \cdot I_{ij} \quad (\text{inertia tensor}) \quad (7)$$

Infinite Homogeneous Fields:

$$\nabla^2 m = 4\pi G \rho_0 m + \Lambda_T m \quad (8)$$

$$\xi_{\text{eff}} = \sqrt{G} \cdot m = \frac{\xi}{2} \quad (\text{cosmic screening}) \quad (9)$$

$$\Lambda_T = -4\pi G \rho_0 \quad (10)$$

Practical Simplification Note

For practical applications: Since all measurements in our finite, observable universe are performed locally, only the **localized spherical field geometry** (first case above) is required:

$\xi = 2\sqrt{G} \cdot m$ and $\beta = \frac{2Gm}{r}$ for all applications.

The other geometries are shown for theoretical completeness but are not needed for experimental predictions.

Natural Units Framework Integration

The complete natural units system where $\hbar = c = \alpha_{\text{EM}} = \beta_T = 1$ provides:

- Universal energy dimensions: All quantities expressed as powers of $[E]$
- Unified coupling constants: $\alpha_{\text{EM}} = \beta_T = 1$ through Higgs physics
- Connection to Planck scale: $\ell_P = \sqrt{G}$ and $\xi = r_0/\ell_P$
- Fixed parameter relationships: No adjustable constants in the theory

.2 Complete Field Equation Framework

Spherically Symmetric Solutions

For a point mass source $\rho = m\delta^3(\vec{r})$, the complete geometric solution is:

$$m(x, t)(r) = m_0 \left(1 + \frac{2Gm}{r} \right) = m_0(1 + \beta) \quad (11)$$

Therefore:

$$T(r) = \frac{1}{m(x, t)(r)} = \frac{1}{m_0} (1 + \beta)^{-1} \approx \frac{1}{m_0} (1 - \beta) \quad (12)$$

Geometric interpretation: The factor 2 in $r_0 = 2Gm$ emerges from the relativistic field structure, exactly matching the Schwarzschild radius.

Modified Field Equation for Infinite Systems

For infinite, homogeneous fields, the field equation requires modification:

$$\nabla^2 m(x, t) = 4\pi G \rho_0 m(x, t) + \Lambda_T m(x, t) \quad (13)$$

where the consistency condition for homogeneous background gives:

$$\Lambda_T = -4\pi G \rho_0 \quad (14)$$

Dimensional verification: $[\Lambda_T] = [4\pi G \rho_0] = [1][E^{-2}][E^4] = [E^2] \checkmark$

This modification leads to the cosmic screening effect: $\xi_{\text{eff}} = \xi/2$.

.3 Lagrangian Formulation with Dimensional Consistency

Time Field Lagrangian Density

The fundamental Lagrangian density for the intrinsic time field is:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu T(x, t) \partial_\nu T(x, t) - V(T(x, t)) \right] \quad (15)$$

Dimensional verification:

- $[\sqrt{-g}] = [E^{-4}]$ (4D volume element)
- $[g^{\mu\nu}] = [E^2]$ (inverse metric)
- $[\partial_\mu T(x, t)] = [E][E^{-1}] = [1]$ (dimensionless gradient)
- $[g^{\mu\nu} \partial_\mu T(x, t) \partial_\nu T(x, t)] = [E^2][1][1] = [E^2]$
- $[V(T(x, t))] = [E^4]$ (potential energy density)
- Total: $[E^{-4}][E^2 + E^4] = [E^{-2}] + [E^0] \checkmark$

(Note of 30 Sept. 2026: The listing gives $[E^{-4}][E^2] = [E^{-2}]$ for the kinetic term and $[E^{-4}][E^4] = [E^0]$ for the potential term – two different dimensions; with these assignments $\mathcal{L}_{\text{time}}$ is not dimensionally uniform, and the check mark is not justified. This also contradicts the later statement that each term has dimension $[E^0]$.)

Modified Schrödinger Equation

The quantum mechanical evolution equation becomes:

$$iT(x, t) \frac{\partial}{\partial t} \Psi + i\Psi \left[\frac{\partial T(x, t)}{\partial t} + \vec{v} \cdot \nabla T(x, t) \right] = \hat{H} \Psi \quad (16)$$

Dimensional verification:

- $[iT(x, t) \partial_t \Psi] = [E^{-1}][E][\Psi] = [\Psi]$

- $[i\Psi\partial_t T(x, t)] = [\Psi][E^{-1}][E] = [\Psi]$
- $[\hat{H}\Psi] = [E][\Psi] = [\Psi] \checkmark$

Higgs Field Coupling

The Higgs field couples to the time field through:

$$\mathcal{L}_{\text{Higgs-T}} = |D_{\text{Higgs-T}}|^2 - V(T(x, t), \Phi) \quad (17)$$

where:

$$D_{\text{Higgs-T}} = T(x, t)(\partial_\mu + igA_\mu)\Phi + \Phi\partial_\mu T(x, t) \quad (18)$$

This establishes the fundamental connection:

$$T(x, t) = \frac{1}{y\langle\Phi\rangle} \quad (19)$$

.4 Matter Field Coupling Through Conformal Transformations

Conformal Coupling Principle

All matter fields couple to the time field through conformal transformations of the metric:

$$g_{\mu\nu} \rightarrow \Omega^2(T(x, t))g_{\mu\nu}, \quad \text{where} \quad \Omega(T(x, t)) = \frac{T_0}{T(x, t)} \quad (20)$$

Dimensional verification: $[\Omega(T(x, t))] = [T_0/T(x, t)] = [E^{-1}]/[E^{-1}] = [1]$ (dimensionless) $\checkmark$

Scalar Field Lagrangian

For scalar fields:

$$\mathcal{L}_\phi = \sqrt{-g}\Omega^4(T(x, t)) \left(\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2 \right) \quad (21)$$

Dimensional verification:

- $[\Omega^4(T(x, t))] = [1]$ (dimensionless)
- $[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi] = [E^2][E^2] = [E^4]$
- $[m^2\phi^2] = [E^2][E^2] = [E^4]$
- Total: $[E^{-4}][1][E^4] = [E^0]$ (dimensionless) $\checkmark$

Fermion Field Lagrangian

For fermion fields:

$$\mathcal{L}_\psi = \sqrt{-g}\Omega^4(T(x, t)) \left(i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi \right) \quad (22)$$

Dimensional verification:

- $[i\bar{\psi}\gamma^\mu\partial_\mu\psi] = [E^{3/2}][1][E][E^{3/2}] = [E^4]$
- $[m\bar{\psi}\psi] = [E][E^{3/2}][E^{3/2}] = [E^4]$
- Total: $[E^{-4}][1][E^4] = [E^0]$ (dimensionless) $\checkmark$

.5 Connection to Higgs Physics and Parameter Derivation

The Universal Scale Parameter from Higgs Physics

The T0 model's fundamental scale parameter, $\xi = 4/30000$, can be checked against Higgs physics. The calculation yields:

$$\xi \approx \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.30 \times 10^{-4} \quad (23)$$

(Corrected on 1 Oct. 2026: previously "uniquely determined through quantum field theory and Higgs physics", $\approx 1.33 \times 10^{-4}$, $\lambda_h \approx 0.13$ – with the SM self-coupling $\lambda_h = m_h^2/(2v^2) \approx 0.129$ one has $\xi = m_h^2/(64\pi^3 v^2) = 1.30 \times 10^{-4}$ ($v = 246$ GeV, $m_h = 125$ GeV), about 2.4 % below $4/30000$; a consistency check of ξ , not an exact derivation; origin in Doc. 385.)

where:

- $\lambda_h = m_h^2/(2v^2) \approx 0.129$ (Higgs self-coupling, dimensionless)
- $v \approx 246$ GeV (Higgs VEV, dimension $[E]$)
- $m_h \approx 125$ GeV (Higgs mass, dimension $[E]$)

Complete dimensional verification:

$$[\xi] = \frac{[1][E^2]}{[1][E^2]} = \frac{[E^2]}{[E^2]} = [1] \quad (\text{dimensionless}) \checkmark \quad (24)$$

Universal Scale Parameter

Key Insight: The parameter $\xi(m) = 2Gm/\ell_p$ scales with mass, revealing the **fundamental unity of geometry and mass**. At the Higgs mass scale, $\xi_0 \approx 1.33 \times 10^{-4}$ provides the natural reference value that characterizes the coupling strength between the time field and physical processes in the T0 model.

Connection to β_T Parameter

The relationship between the scale parameter and the time field coupling is established through:

$$\beta_T = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2 \xi} = 1 \quad (25)$$

This relationship, combined with the condition $\beta_T = 1$ in natural units, links ξ to the Higgs sector; with $\xi = 4/30000$ and the SM values it gives $\beta_T \approx 0.976$, i.e. consistency to about 2.4 %. (Corrected on 1 Oct. 2026: previously "uniquely determines ξ and eliminates all free parameters from the theory" – $\beta_T = 1$ is a setting; with it ξ would only be defined by the Higgs expression, which is circular; with $\lambda_h = m_h^2/(2v^2)$ one has $m_h^2/(64\pi^3 v^2 \xi) = 0.976$ for $\xi = 4/30000$.)

Geometric Modifications for Different Field Regimes

The universal scale parameter ξ undergoes geometric modifications depending on the field configuration:

- **Localized fields:** $\xi = 1.33 \times 10^{-4}$ (full value)
- **Infinite homogeneous fields:** $\xi_{\text{eff}} = \xi/2 = 6.7 \times 10^{-5}$ (cosmic screening)

This factor of $1/2$ reduction arises from the Λ_T term in the modified field equation for infinite systems and represents a fundamental geometric effect rather than an adjustable parameter.

.6 Complete Total Lagrangian Density

Full T0 Model Lagrangian

The complete Lagrangian density for the T0 model is:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{time}} + \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\phi} + \mathcal{L}_{\psi} + \mathcal{L}_{\text{Higgs-T}} \quad (26)$$

where each component is dimensionally consistent:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_{\mu} T(x, t) \partial_{\nu} T(x, t) - V(T(x, t)) \right] \quad (27)$$

$$\mathcal{L}_{\text{gauge}} = \sqrt{-g} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \quad (28)$$

$$\mathcal{L}_{\phi} = \sqrt{-g} \Omega^4(T(x, t)) \left(\frac{1}{2} g^{\mu\nu} \partial_{\mu} \phi \partial_{\nu} \phi - \frac{1}{2} m^2 \phi^2 \right) \quad (29)$$

$$\mathcal{L}_{\psi} = \sqrt{-g} \Omega^4(T(x, t)) \left(i \bar{\psi} \gamma^{\mu} \partial_{\mu} \psi - m \bar{\psi} \psi \right) \quad (30)$$

$$\mathcal{L}_{\text{Higgs-T}} = \sqrt{-g} |D_{\text{Higgs-T}}|^2 - V(T(x, t), \Phi) \quad (31)$$

Dimensional consistency: Each term has dimension $[E^0]$ (dimensionless), ensuring proper action formulation.

.7 Cosmological Applications

Modified Gravitational Potential

The T0 model predicts a modified gravitational potential:

$$\Phi(r) = -\frac{GM}{r} + \kappa r \quad (32)$$

where κ depends on the field geometry:

- **Localized systems:** $\kappa = \alpha_{\kappa} H_0 \xi$
- **Cosmic systems:** $\kappa = H_0$ (Hubble constant)

Static Universe Interpretation

The T0 model explains cosmological observations without spatial expansion:

- **Redshift:** Energy loss to time field gradients
- **Cosmic microwave background:** Equilibrium radiation in static universe
- **Structure formation:** Gravitational instability with modified potential
- **Dark energy:** Emergent from Λ_T term in field equation

.8 Experimental Predictions and Tests

Distinctive T0 Signatures

The T0 model makes specific testable predictions using the universal scale parameter $\xi \approx 1.33 \times 10^{-4}$:

1. Wavelength-dependent redshift:

$$\frac{z(\lambda_2) - z(\lambda_1)}{z_0} = \ln \frac{\lambda_2}{\lambda_1} \quad (33)$$

2. QED corrections to anomalous magnetic moments:

$$a_t^{(T0)} = \frac{\alpha}{2\pi} \xi^2 I_{\text{loop}} \approx 2.3 \times 10^{-10} \quad (34)$$

3. Modified gravitational dynamics:

$$v^2(r) = \frac{GM}{r} + \kappa r^2 \quad (35)$$

4. Energy-dependent quantum effects:

$$\Delta t = \frac{\xi}{c} \left(\frac{1}{E_1} - \frac{1}{E_2} \right) \frac{2Gm}{r} \quad (36)$$

Precision Tests

The fixed-parameter nature allows stringent tests:

- **No free parameters:** All coefficients derived from $\xi \approx 1.33 \times 10^{-4}$
- **Cross-correlation:** Same parameters predict multiple phenomena
- **Universal predictions:** Same ξ value applies across all physical processes
- **Quantum-gravitational connection:** Tests of unified framework

.9 Dimensional Consistency Verification

Complete Verification Table

Equation	Left Side	Right Side	Status
Time field definition	$[T] = [E^{-1}]$	$[1/\max(m, \omega)] = [E^{-1}]$	✓
Field equation	$[\nabla^2 m] = [E^3]$	$[4\pi G \rho m] = [E^3]$	✓
β parameter	$[\beta] = [1]$	$[2Gm/r] = [1]$	✓
ξ parameter (Higgs)	$[\xi] = [1]$	$[\lambda_h^2 v^2 / (16\pi^3 m_h^2)] = [1]$	✓
β_T relationship	$[\beta_T] = [1]$	$[\lambda_h^2 v^2 / (16\pi^3 m_h^2 \xi)] = [1]$	✓
Energy loss rate	$[dE/dr] = [E^2]$	$[g_T \omega^2 2G/r^2] = [E^2]$	✓
Modified potential	$[\Phi] = [E]$	$[GM/r + \kappa r] = [E]$	✓
Lagrangian density	$[\mathcal{L}] = [E^0]$	$[\sqrt{-g} \times \text{density}] = [E^0]$	✓
QED correction	$[a_t^{(T0)}] = [1]$	$[\alpha \xi^2 / 2\pi] = [1]$	✓

Table 1: Complete dimensional consistency verification for T0 model equations

.10 Connection to Quantum Field Theory

Modified Dirac Equation

The Dirac equation in the T0 framework becomes:

$$[i\gamma^\mu(\partial_\mu + \Gamma_\mu^{(T)}) - m(x, t)]\psi = 0 \quad (37)$$

where the time field connection is:

$$\Gamma_\mu^{(T)} = \frac{1}{T(x, t)} \partial_\mu T(x, t) = -\frac{\partial_\mu m}{m} \quad (38)$$

(Corrected on 30 Sept. 2026: previously $-\partial_\mu m/m^2$ – with $T(x, t) = 1/m$ one has $\partial_\mu T(x, t) = -\partial_\mu m/m^2$, hence $(1/T(x, t)) \partial_\mu T(x, t) = m \cdot (-\partial_\mu m/m^2) = -\partial_\mu m/m$.)

QED Corrections with Universal Scale

The time field introduces corrections to QED calculations using the universal scale parameter:

$$a_e^{(T0)} = \frac{\alpha}{2\pi} \cdot \xi^2 \cdot I_{\text{loop}} = \frac{1}{2\pi} \cdot (1.33 \times 10^{-4})^2 \cdot \frac{1}{12} \approx 2.34 \times 10^{-10} \quad (39)$$

This prediction applies universally to all leptons, reflecting the fundamental nature of the scale parameter.